Real-Estate Price Prediction: From-Scratch Trees and Pegasos SVMs

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***Abstract*—**We study Azerbaijani real-estate listings from bina.az through two supervised tasks: predicting price in AZN and classifying listings as standard or premium relative to the training median. The pipeline cleans mixed-type fields, removes target-derived leakage, encodes categoricals, and fits imputation and scaling using training data only. We implement a NumPy decision tree and a Pegasos soft-margin SVM, with Random Fourier Features (RFF) as a nonlinear extension, and compare them with scikit-learn baselines. On the held-out test split, the scratch tree obtains RMSE 380,596 AZN and $R^2 = 0.688$ for regression; the scratch linear Pegasos model obtains F1 0.826 and ROC-AUC 0.911. The scikit-learn tree gives the highest classification F1 (0.930) and ROC-AUC (0.967). Linear Pegasos remains competitive with linear baselines, whereas RFF is sensitive to bandwidth and finite feature dimension. Code is available at `github.com/Haji708/ml_final_project`.

***Index Terms*—**decision tree, support-vector machine, Pegasos, regression, classification, real-estate price prediction.

I. Introduction

Online property markets contain heterogeneous numeric, categorical, and geospatial signals. We frame the problem as Task A, price regression, and Task B, binary price-tier classification. Our contributions are a shared leakage-aware preprocessing pipeline, NumPy implementations of a decision tree and Pegasos SVM, an RFF extension, and matched scikit-learn benchmarks.

II. Related Work / Background

The tree recursively chooses numeric thresholds that maximize weighted impurity reduction. Gini impurity and entropy are used for classification, while variance/MSE reduction is used for regression. Leaves predict the majority class and class frequencies, or the mean target. The SVM minimizes the regularized hinge-loss primal with stochastic subgradient updates. Pegasos avoids a quadratic-programming solver and uses $\eta_t = 1/(\lambda t)$ [1]. RFF approximates an RBF feature map before applying the same linear optimizer.

III. Data and Preprocessing

The downloaded bina.az snapshot contains 100,775 raw listings. We enforce AZN prices, coerce numeric strings, parse mixed units, and remove 34 records outside the fixed eligibility range $1{,}000 \le \text{price} \le 20{,}000{,}000$ AZN. Exact duplicates and duplicate identifiers were absent. Missing numeric values use training-fitted medians and missingness indicators; categorical values are fitted from the training partition. The resulting modelling dataset has 100,705 rows and 178 encoded features.

Task A predicts price and uses RMSE, MAE, and R^2. Task B defines premium as price above the training median of 218,000 AZN and standard otherwise; it uses precision, recall, F1, and ROC-AUC. The deterministic seed-42 split is 70,493/15,106/15,106 train/validation/test rows, sampled within price-rank strata. This balances price ranks but is not exact binary tier stratification. The leakage columns `unit_price` and `total_price` are removed because they reveal or derive the target. IDs, URLs, owner names, addresses, and free text are also excluded.

Table 1. Dataset summary after cleaning.

Property	Value
Rows (after cleaning)	100,705
Features used	178 encoded features
Train / Val / Test split	70,493 / 15,106 / 15,106
Training tier balance	50.14% / 49.86%

IV. Method

A. Decision tree (from scratch)

For each candidate feature, values are sorted and thresholds are tested between distinct adjacent values. The selected split maximizes parent impurity minus the sample-weighted child impurities. Growth is recursive and stops at maximum depth, insufficient samples, invalid child sizes, or non-positive impurity reduction. Classification leaves store class probabilities and predict the majority class; regression leaves predict the target mean. Validation experiments use depth and minimum-leaf controls to limit overfitting, with Gini, entropy, or MSE selected according to the task.

B. Support-vector machine (from scratch)

The objective and Pegasos update are:

$$J(\mathbf{w}, b) = \tfrac{\lambda}{2}\|\mathbf{w}\|^2 + \tfrac{1}{n}\sum_{i=1}^{n} \max\big(0,\, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\big). \quad (1)$$

Labels are mapped to $\{-1, +1\}$. At step t, a sampled example with margin $m_i = y_i(\mathbf{w}^\top \mathbf{x}_i + b) < 1$ updates both weights and bias; otherwise only regularization shrinkage is applied to $\mathbf{w}$. The implementation records the objective during fitting. All SVM inputs are standardized using training means and scales only. The nonlinear extension uses RFF:

$$\phi(\mathbf{x}) = \sqrt{2/D}\cos(\mathbf{x}W + \mathbf{r}),$$

where W and $\mathbf{r}$ are generated once during fitting and reused at prediction time. Support vectors satisfy $y_i f(\mathbf{x}_i) \le 1 + 10^{-6}$, and the geometric margin width is $2/\|\mathbf{w}\|$.

Table 2. Headline results on the held-out test set.

Model	Task A (RMSE)	Task B (F1)
Ours: decision tree	380,596	0.941
Ours: SVM (Pegasos)	n/a	0.826
scikit-learn baseline	401,341	0.930

V. Experimental Setup

We report RMSE, MAE, and R^2 for Task A, and precision, recall, F1, ROC-AUC, and confusion-matrix diagnostics for Task B. Hyperparameters are selected on validation data only. The linear Pegasos sweep tests $\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}\}$; RFF trials vary bandwidth and component count. The selected linear model uses $\lambda = 10^{-3}$, 50,000 updates, and the fixed seed is 42. Baselines include normal-equation Ridge, scikit-learn Ridge, DecisionTreeRegressor/Classifier, SGDClassifier with hinge loss, and LinearSVC.

VI. Results

Lead with a results table; back every claim with a number or figure.

VII. Discussion and Limitations

The scratch tree gives the best reported regression test performance (RMSE 380,596 AZN, MAE 71,856 AZN, $R^2 = 0.688$), improving on the scikit-learn tree's RMSE 401,341 AZN. For classification, the scikit-learn tree is strongest overall (F1 0.930, ROC-AUC 0.967), while linear Pegasos reaches F1 0.826 and ROC-AUC 0.911. Its result is close to SGDClassifier (F1 0.838) and LinearSVC (F1 0.835), supporting the correctness of the from-scratch linear solver.

RFF is a valid nonlinear extension but is sensitive to bandwidth, random approximation error, and component count. On this snapshot its test F1 is 0.832 and ROC-AUC is 0.908, so the finite approximation did not improve on the linear boundary. Large residuals are expected for luxury listings and sparse locations; a single global model cannot fully represent local market effects. The fixed split is not cross-validated, so these values quantify this snapshot rather than uncertainty across folds. Future work should add repeated or temporal validation, richer geospatial features, and systematic RFF bandwidth tuning.

VIII. Conclusion

The project demonstrates a complete leakage-aware workflow for bina.az price prediction and price-tier classification. From-scratch Pegasos matches the expected hinge-loss formulation and remains competitive with linear scikit-learn baselines, while the custom tree captures stronger nonlinear structure in this dataset. The next priority is repeated or time-based validation and fuller analysis of regional and luxury-listing errors.

Tools and Acknowledgements

The team used NumPy, pandas, matplotlib, and scikit-learn for data processing, modelling, evaluation, and plotting. An AI coding assistant was used for code navigation, debugging suggestions, LaTeX editing, and test-oriented review; all generated changes were inspected and validated by the team. The raw bina.az data were downloaded from the Kaggle `binaaz-sale-project` dataset by Sehriyar Memmedli and are not redistributed in this repository.

References

[1] S. Shalev-Shwartz, Y. Singer, N. Srebro, and A. Cotter, "Pegasos: primal estimated sub-gradient solver for SVM," *Mathematical Programming*, vol. 127, no. 1, pp. 3–30, 2011.

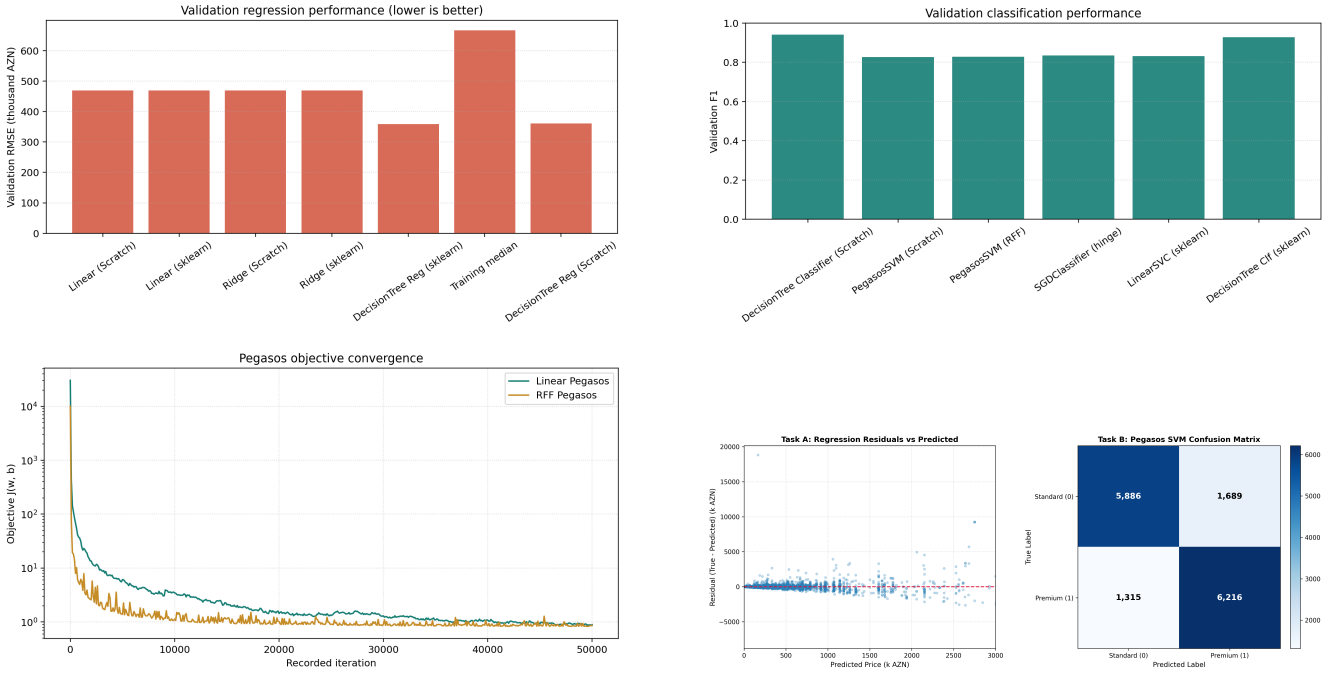


Figure 1. Validation performance, Pegasos objective convergence, and test error analysis. Regression RMSE is lower-is-better; classification F1 is higher-is-better.